# Attorney Directory and Case File Management System

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# CHAPTER I

# INTRODUCTION

## 1.1 Background of the Study

As the industry and workforce continue to evolve and compete by entering the digital bubble, the demand for larger, improved forms of storage becomes essential. Public attorney offices are no exception—offices in nature are notorious for keeping large cabinets filled with cases of different statuses, be it ongoing or finished. The demand continues to rise and the dire need for space becomes a concern. Bulky cabinets and drawers that are volatile to damage such as termites, unpredictable weather conditions, and the wear of time put an archive at risk. Attorneys meet with countless clients, whilst having to face a multitude of cases at a time. Attorneys also tend to leave case files nearby—either at their desks or inside their cabinets. While others may argue that it becomes second nature for them to keep track of all the ongoing cases, keeping track of cases of utmost priority versus those that have been solved can be tedious as the workload increases. Thus, not only is an easily accessible system prevalent, but safer storage to keep archives intact for time to come and potential future record verification is urgently needed.

Furthermore, retrieving legal documentation and connecting with qualified legal professionals remains a slow, paper-heavy process. While legal researchers or authorized external parties may need to verify attorney credentials or request public, non-confidential court records, acquiring these documents from private offices involves tedious administrative processes that

take up valuable time. On the client side, individuals often must rely on word-of-mouth or conduct intensive research just to find active, qualified attorneys to handle their specific cases.

Hence, the creation of the **Attorney Directory and Case File Archive System**. The system brings together a secure, cloud-synchronized archive to protect physical case records from environmental damage and streamline document retrieval, alongside a public-facing directory where clients and authorized users can search verified attorney profiles, view specializations, and access public case histories directly linked from the internal archive—all while maintaining strict role-based access to safeguard confidential attorney-client records.

## 1.2 Statement of the Problem

As stated in the background, bulky storage can affect case files that are kept for months and years, leaving cases vulnerable to physical damage and wear: termites, weather conditions, natural disasters, and heat can all be factors on ruining the cases inside the archives; it can start to be tedious to keep track of the cases of their respective priorities.

Moreover, cases and legal records tend to be locked behind the offices of attorneys, making file retrieval slow and difficult. While law enforcement personnel or legal researchers may need to verify an attorney's credentials or request public, non-confidential court records, acquiring these documents from private offices involves tedious processes that take up valuable time. On the other side, prospective clients have to follow word of mouth or do intensive research just to look for qualified attorneys to handle their cases.

Hence, the creation of the Attorney Directory and Case File Archive System. The system serves a dual purpose: the Case File Archive acts as the internal tool for attorneys to safely store, organize, and quickly retrieve their records, while the Attorney Directory acts as the public-facing hub where clients and authorized users can search verified attorney profiles, view their specializations, and access public case histories directly linked from the archive.

## 1.3 Objectives of the Study

The Attorney Directory and Case File Archive System aims to provide these objectives (general and specific objectives):

### 1.3.1 General Objectives:

Design, develop, and implement a web-based Attorney Directory and Case File Archive System that digitalizes and centralizes legal records to prevent physical damage to case files, streamline file retrieval, and provide a structured, role-based access to case histories while creating a centralized hub for clients, legal researchers, and legal professionals to look up verified attorney profiles.

### 1.3.2 Specific Objectives:

1. To design a digital archiving module capable of uploading, categorizing, and converting physical case documents into secure, versatile digital formats.
2. To implement an automated cloud-based backup and recovery protocol that ensures data redundancy and protects case archives against physical disasters, data loss, or system failures.
3. To develop an advanced search and indexing tool that allows users (based on permission level) to quickly retrieve archived cases using metadata parameters.
4. To integrate a priority tracking and classification dashboard that helps legal administrators monitor active, pending, and closed cases according to their urgency or court schedules.
5. To build a public-facing Attorney Directory that enables potential clients to search, view profiles, evaluate legal credentials, and directly contact active attorneys.

6. To establish a secure, controlled portal for public cases that allows clients and authorized users to view historical, non-confidential case files directly linked from the archive without needing physical office access, while maintaining strict role-based access to safeguard confidential attorney-client records.

## 1.4 Significance of the Study

The findings and implementation of this study aim to provide a practical technological solution to traditional, paper-heavy legal case management and attorney verification processes. Specifically, this study will be beneficial to the following stakeholders:

- **Legal Practitioners and Attorneys:** This study provides attorneys with a centralized, secure digital repository that protects sensitive case records against environmental degradation, physical loss, and unauthorized access. It streamlines daily operations through metadata indexing and priority tracking dashboards, allowing attorneys to manage caseloads efficiently and monitor critical court schedules.
- **Law Office Administrative Staff (Legal Assistants and Clerks):** The system reduces the manual administrative burden of organizing physical dockets, tracking case statuses, initializing client profiles, and handling routine inquiries. By digitizing the archiving process, staff can quickly locate files, schedule court hearings, and manage record updates with minimal operational friction.
- **Prospective Clients and the Public:** The public-facing Attorney Directory offers clients a transparent, convenient hub to find qualified legal professionals, evaluate specializations, and verify credentials. Furthermore, clients benefit from the public case portal, which allows them to view non-confidential historical records remotely without needing to travel to physical offices.

- **Legal Researchers and Academics:** This study offers researchers a structured and accessible platform for looking up non-confidential court disposition histories and verified legal profiles, significantly reducing the time spent on manual document retrieval during legal studies.

## 1.5 Scopes and Delimitations

The system focuses on mitigating the time needed to access legal files while protecting case records against physical damage and wear caused by environmental factors, natural disasters, or improper storage. It provides a web-based platform that centralizes case archiving and public attorney verification for legal professionals, clients, and authorized external users.

The system allows attorneys and authorized office staff to create and update client profiles, initialize case records, update case statuses, attach digital documents (e.g., filings, court orders), and organize records within their respective case lists. It features automated document versioning, immutability tracking, an append-only audit log, and a priority lifecycle dashboard to help monitor active, pending, closed, and archived cases based on urgency or scheduled court dates. Archived cases are automatically synchronized with an encrypted cloud-based backup system to ensure data redundancy, disaster recovery, and long-term retention.

Furthermore, the system includes an advanced metadata search tool that allows users, based on their assigned permission levels, to quickly locate specific records. It also provides a public-facing Attorney Directory where prospective clients and authorized users can look up verified attorney profiles, credentials, and contact details, as well as view historical, non-confidential case files directly linked from the archive without needing physical office access. Strict role-based access controls are enforced to protect private attorney-client records and ensure compliance with data privacy standards.

The system is delimited to case file archiving and attorney directory management. It does not handle automated legal advice, AI-driven case evaluations, or drafting of legal decisions. The system does not integrate with official Philippine court e-filing platforms (such as eCourt PH) or government agency dockets. Additionally, the system excludes financial management, legal fees, invoices, online payment processing, and real-time video or chat communications. The physical scanning of paper documents is also excluded, as the system's functions begin once digital files are uploaded into the database

## 1.6 Definition of Terms

- **AES-256 (Advanced Encryption Standard):** A symmetric encryption algorithm utilizing a 256-bit key to encrypt stored digital files in AWS S3, ensuring military-grade confidentiality for archived legal records.
- **Atomic Transaction:** An indivisible sequence of operations where cloud file storage and database updates execute together as an all-or-nothing process, automatically rolling back changes if any step fails to prevent orphaned records.
- **Attorney Directory:** A centralized, public-facing digital registry within the system that displays verified attorney profiles, credentials, specializations, and contact information to help potential clients find and hire legal representation.
- **AWS S3 (Amazon Web Services Simple Storage Service):** An off-site cloud object storage infrastructure used by the system to hold encrypted binary document attachments with automated versioning and lifecycle policies.
- **Bcrypt:** A secure password-hashing function that uses adaptive hashing and salting to store user account passwords as non-reversible cryptographic strings in the database.
- **Case File Archive:** The secure digital repository within the system where historical and active case documents, metadata, and judicial histories are preserved, eliminating the need for bulky physical paper storage.

- **Case Priority**: A classification system (e.g., High, Medium, Low) assigned to case files to help attorneys and administrators track critical deadlines, urgent hearings, and prioritized legal tasks.
- **Case Status:** The current state of legal proceedings stored in the database (e.g., Active, Pending, Closed, Archived), which determines its visibility and modification permissions within the system.
- **Client Registry:** A structured compilation of personal and demographic information of individuals seeking legal counsel, directly linked to their corresponding case files and assigned attorneys.
- **Digital Preservation:** The process of scanning, converting, and storing physical legal records into versatile, search-friendly digital formats (like PDFs) to protect them from environmental hazards such as termites, weather, and physical wear.
- **Immutable Audit Trail:** An unalterable, system-generated log that automatically captures user actions, security events, and record modifications for administrative accountability and security compliance.
- **JWT (JSON Web Token):** A compact, stateless security standard used to authenticate API requests and maintain user session permissions securely across the web application.
- **Lifecycle Retention Policy:** An automated data governance configuration within cloud storage that retains historical legal revisions for a mandatory 7-year duration prior to permanent disposition.

- **Multer:** A server-side Node.js middleware package used to intercept incoming file uploads, enforce a 25 MB file size limit, and validate PDF/DOCX file formats before cloud transmission.
- **Pre-Signed URL:** A temporary, time-limited cryptographic web link generated by the system to grant authorized users short-term, secure access to private cloud documents.
- **RBAC (Role-Based Access Control):** An authorization framework that restricts system features and document access based on assigned user roles (Public, Staff, Attorney, Admin).
- **Soft-Delete:** A database mechanism where records are flagged as inactive (is_deleted = TRUE) to hide them from the primary interface while preserving historical records in MySQL.

# CHAPTER 2

# REVIEW OF RELATED LITERATURE AND STUDIES

## 2.1 Related Literature

### 2.1.1 Foreign Literature

Kiršienė et al. (2022) evaluate the global shift toward digital legal transformation, demonstrating that public-facing portals, online directories, and structured digital communication tools significantly expand access to legal services while reducing transactional friction inherent in traditional legal practice. The authors emphasize that integrated digital platforms remove geographical and administrative barriers for citizens seeking legal assistance. However, while the study effectively illustrates the broad societal benefits of public digital access, it focuses primarily on high-level administrative concepts and offers limited technical insight into how public-facing directory services and confidential legal record repositories can coexist within a unified web environment without compromising operational security.

Swathilakshmi et al. (2026) examine emerging legal technology architectures, emphasizing that secure web platforms provide transparent operational frameworks that demystify legal processes, lower information asymmetries, and build public trust in administrative workflows. While the research highlights the necessity of structural transparency and open communication between legal professionals and clients, the authors concentrate heavily on theoretical software models. Consequently, the study

leaves a gap regarding how localized, resource-constrained web platforms can maintain seamless relational data tracking and transparent status updates under routine administrative use.

In a global study on judicial and administrative modernization, the World Bank (2023) establishes that replacing physical paper registries with electronic case tracking architectures resolves severe administrative backlogs, enhances processing speed, and enforces operational transparency across institutional workflows. Nevertheless, because the World Bank report focuses predominantly on large-scale national judicial systems backed by substantial state budgets, it does not fully address the specific deployment constraints, structural storage limits, or localized data management requirements encountered by independent legal practices and regional archiving platforms.

Zeleznikow (2021) examines intelligent information retrieval within legal platforms, highlighting that automated workflow indexing and structured data processing minimize human discretion, accelerate case disposition times, and ensure consistent application of procedural rules across administrative bodies. However, the analysis centers primarily on algorithmic decision-making and automated rule engines, giving less attention to foundational administrative requirements such as multi-user access tracking, historical action logging, or the systematic management of multi-version static document uploads.

Wilson-Kovacs (2021) examines the operational challenges faced by digital media investigators, demonstrating that managing massive quantities of digital records create severe processing bottlenecks. Her work underscores the necessity of implementing standardized organizational management tools, clear file indexing protocols, and secure inter-departmental collaboration channels to maintain administrative capacity. While the analysis focuses strictly on investigative operations and evidence preservation— leaving out public-facing administrative features such as directory lookup or client appointment management— it underscores the vital importance of structured file indexing to prevent digital record congestion over time.

### 2.1.2 Local Literature

Bañez (2025) analyzes the ongoing digitization of the Philippine justice system, demonstrating that updating procedural rules alongside electronic filing and automated document routing eliminates physical record degradation, mitigates document loss, and enhances overall transparency across legal proceedings. The study emphasizes that procedural modernization must accompany technological adoption to prevent administrative bottlenecks in Philippine legal practice. However, while the paper provides an insightful analysis of procedural rules and legal policy shifts, its evaluation centers primarily on high-level judicial reform. It offers limited technical evaluation regarding how localized, web-accessible platforms can maintain secure, multi-party access control when handling both public directory searches and confidential legal record repositories.

The Department of Information and Communications Technology (DICT, 2023) establishes in the E-Government Masterplan 2023–2028 that modernizing government operations requires transitioning physical agency registries into integrated, web-accessible digital frameworks to ensure efficient service delivery across public bodies. The masterplan highlights interoperability, secure data sharing, and digital accessibility as fundamental pillars for public sector efficiency in the Philippines. Nevertheless, because the document serves as a high-level strategic roadmap for national agency compliance, it lacks specific architectural guidelines for independent legal bodies or localized archival

environments that must operate within strict data privacy constraints while providing open public directory services.

The National Economic and Development Authority (NEDA, 2023) outlines in the Philippine Development Plan 2023–2028 that seamless digital governance is essential for strengthening public trust, eliminating bureaucratic friction, and boosting administrative performance. NEDA emphasizes that replacing manual, paper-reliant filing systems with secure management portals directly addresses severe processing delays and streamlines public verification requests. While the policy framework strongly justifies the societal and economic necessity of public sector digitization, it does not address the granular operational challenges of legal archiving—such as multi-version document control, attorney credential verification, or preserving the chain of custody for archived case materials.

Dela Gente et al. (2022) examine administrative performance and organizational workflows in Philippine public and private sectors, proving that eliminating rigid, manual documentation steps significantly reduces administrative friction and fosters an environment supportive of modern, innovative work practices. The study demonstrates that digital record accessibility directly improves staff productivity and reduces turnaround times for public requests. However, the authors focus broadly on general administrative office environments rather than specialized legal settings, leaving a gap regarding how administrative platforms can balance

operational speed with the strict confidentiality and regulatory compliance required in legal record management.

The Supreme Court of the Philippines (2022) establishes in its Strategic Plan for Judicial Innovations 2022–2027 (SPJI) that public legal administration must leverage modern information technology to deliver transparent, accessible, and timely legal services. The framework mandates the adoption of secure, role-based document archives and client-facing digital tracking tools to modernize case management workflows, safeguard record confidentiality, and bridge the access gap between legal institutions and citizens. While the SPJI provides the ultimate institutional directive for legal technology in the country, its focus remains primarily on the formal court system, leaving independent law offices, local practitioner directories, and private case repositories to develop compatible web-based solutions that align with these national judicial standards.

## 2.2 Related Studies

### 2.2.1 Foreign Studies

Skipanes et al. (2026) conducted an empirical investigation examining the deployment of Case Management Systems (CMS) in criminal inquiries. Their study proves that structured digital repositories and automated file indexing significantly reduce cognitive load on investigators while preventing fragmented departmental silos from obscuring vital evidence. However, while the study effectively highlights the operational benefits of structured case repositories, its empirical focus centers strictly on investigative law enforcement environments. Consequently, it leaves a gap regarding how open-access web portals can safely balance public practitioner searches with restricted back-end case record management. This study provides empirical support for the proposed system, demonstrating that structured file indexing and centralized repository architecture are essential for preventing data fragmentation and streamlining document retrieval.

James et al. (2025) empirically evaluated operational constraints in legal administrative environments, revealing that high caseloads and unintegrated paper file management force personnel to spend disproportionate hours on administrative logging rather than core duties. Their findings establish that reliance on manual, paper-based record-keeping leads to severe administrative bottlenecks, delayed case processing, and document loss. While the study convincingly quantifies the latency and inefficiency inherent in

paper-based workflows, its scope is limited to diagnosing operational gridlock without testing specific web-based software configurations or software architectures to solve these issues. This study offers strong empirical justification for the proposed system, confirming that replacing manual paper registries with a centralized digital archive directly eliminates administrative latency.

Guttman et al. (2022) published a technical empirical study through the National Institute of Standards and Technology (NIST) investigating the security, authenticity, and long-term integrity of digitized legal files. The authors establish that electronic case files face distinct security vulnerabilities, including metadata loss, silent corruption, and unauthorized alterations. Their technical evaluations demonstrate that digital repositories must incorporate rigorous safeguards—such as immutable audit logging, strict role-based access controls, and standardized metadata tagging—to ensure long-term record preservation and legal admissibility. Although the study provides comprehensive security benchmarks, its analysis focuses primarily on high-level enterprise and institutional archival standards, offering limited evaluation for lightweight web applications. Nevertheless, its findings reinforce the critical necessity of embedding role-based access controls and audit logging mechanisms into the proposed system's case archive architecture.

Munaweerage (2025) designed and empirically evaluated an online legal support platform prototype, demonstrating that centralized web portals offering real-time status updates, electronic document submission, and automated query routing drastically reduce case tracking delays and improve

client-practitioner communication. However, user testing during the study revealed a notable limitation in the platform's long-term utility, as it focused heavily on initial client support interaction while offering limited back-end case lifecycle archiving and multi-party document tracking. This study highlights the operational gaps of single-purpose client portals, directly supporting the development of an integrated platform that unifies client-facing interaction tools with comprehensive back-end case file management.

Punitha et al. (2024) evaluated a web-based lawyer consultancy platform using empirical user testing, proving that verified practitioner profiles, automated appointment scheduling, and direct communication channels lower transaction friction and simplify public engagement with legal professionals. The research established that public trust and user engagement increase significantly when practitioner credentials are clearly verified prior to scheduling interactions. While the study successfully validates the usability of public consultancy tools, its scope remains restricted to legal matchmaking and appointment booking, omitting back-end legal case archiving and multi-version document control. This study directly validates the inclusion of a verified Attorney Directory and scheduling features within the proposed system, demonstrating that public directory verification improves client engagement.

### 2.2.2 Local Studies

Innovations for Poverty Action (IPA, 2023) conducted an institutional empirical evaluation across Philippine trial courts to assess the impact of court management technologies on judicial efficiency. Their study proves that implementing electronic court dockets significantly reduces average case processing times, alleviates severe court congestion, and restores operational accountability across local jurisdictions. However, while the evaluation provides strong quantitative data on macro-level judicial performance, its scope centers on formal state court systems. It offers limited evaluation regarding how regional or localized law practices can implement similar electronic tracking mechanisms under localized web hosting constraints. This study offers strong empirical justification for replacing manual tracking, confirming that digitizing legal registries directly reduces processing delays and operational backlogs in the Philippine context.

Acoba (2024) developed and evaluated the Judicial Court Information Management System (J-CIMS) for a Philippine Regional Trial Court (RTC) using usability testing and system performance evaluations. The study established that centralizing trial court records into a secure web repository eliminates file retrieval delays, prevents document misplacement, and streamlines case scheduling. While J-CIMS successfully addresses record management for trial court personnel, the system was designed exclusively for internal judicial staff, leaving out public-facing directory features, practitioner verification, and client scheduling tools. This research directly supports the development of a

secure case archiving architecture, demonstrating that centralized digital repositories solve file retrieval delays in local legal settings.

Daluyon (2024) designed and tested an electronic document archiving system for local public administrative workflows in the Philippines, utilizing user acceptance metrics and retrieval latency tests. The empirical results proved that automated file indexing, role-based security access, and structured retention schedules drastically improve record retrieval speeds while maintaining compliance with national data privacy standards. However, the study focused on general municipal administrative documents rather than legal case files, leaving a gap regarding the specific workflow requirements of legal practices, such as hearing schedules, attorney assignment, and multi-party case tracking. This study validates the technical necessity of structured file indexing and role-based access control to ensure fast and secure document retrieval.

Antenor and Pomperada (2025) evaluated the implementation of the Docuware e-document management system within a Philippine institutional setting, measuring staff productivity and workflow satisfaction. Their empirical findings demonstrated that transitioning from physical logbooks to digital repositories eliminates manual paper tracking bottlenecks and earns exceptionally high usability ratings among administrative staff. Nevertheless, because Docuware is an off-the-shelf, generalized enterprise document platform, the study highlights the platform's high implementation costs and its lack of tailored legal features, such as public attorney directories or

automated hearing management. This research reinforces the operational value of replacing physical logbooks with web-based archives to boost administrative efficiency and user satisfaction.

Peñalba and Macabeo (2022) analyzed administrative case processing within the Philippine National Police (PNP PRO V) using descriptive-evaluative empirical methods. Their investigation revealed that reliance on manual record-keeping, unindexed physical files, and paper tracking logs leads to severe disposition delays, missing records, and significant administrative case backlogs. The authors concluded that manual systems create severe operational gridlock and emphasize the urgent necessity of deploying centralized, web-based legal case tracking portals to enforce transparent monitoring and accelerating case disposition. This study offers compelling local empirical evidence on the risks of physical record management, directly justifying the deployment of a centralized, web-based case file archive to prevent administrative delays.

## 2.3 Related Systems

### 2.3.1. eCourt PH (Philippine Judiciary Platform)

Developed by the Supreme Court of the Philippines, eCourt PH serves as the official judicial management platform designed to digitize court dockets, electronic case filings, and status tracking across first- and second-level trial courts nationwide. Major features of the platform include automated case assignments to trial judges, electronic pleading submissions, case status dashboards, and digital document tracking for court personnel.

Built on a cloud-based enterprise web architecture with secure database repositories and strict role-based access protocols, the system offers significant advantages by centralizing judicial dockets, reducing manual administrative delays, and providing real-time tracking across jurisdictions. However, its primary disadvantage is its restricted ecosystem; it is built exclusively for internal court staff and active litigants, lacking public-facing capabilities such as a searchable attorney directory or consultation scheduling. In relation to the proposed study, eCourt PH provides a proven operational model for secure, web-based case tracking and digital filing in the Philippines, though the proposed system addresses broader administrative needs by integrating a public attorney lookup alongside secure case archiving.

### 2.3.2. Clio Legal Practice Management

Clio, developed by Themis Solutions Inc., is a cloud-based legal practice management platform designed to streamline law firm workflows, client intake, document handling, and billing operations. Its major features include dedicated client portal, automated document management, appointment scheduling, task tracking, and encrypted file archiving. The system utilizes a Software-as-a-Service (SaaS) cloud infrastructure with RESTful APIs, multi-tenant database management, and end-to-end data encryption. The primary advantages of Clio reside in its comprehensive feature set, high security standards, and intuitive cross-platform accessibility. Conversely, its main disadvantages are its subscription-based commercial model—which carries high recurring costs—and its lack of alignment with Philippine administrative legal standards or public regulatory framework requirements. In relation to the proposed study, Clio illustrates how modern web applications manage law practice workflows and document archives, serving as a functional blueprint for adapting client scheduling and file repositories into a localized, cost-effective platform.

### 2.3.3 MyCase Legal Practice Management

MyCase, developed by AffiniPay, is a cloud-based legal practice management software designed to centralize small and mid-sized law firm operations into a unified digital workspace. Its primary features encompass a secure client portal, automated client intake, document management and file archiving, centralized calendar tracking, and integrated billing workflows. The system leverages a multi-tenant Software-as-a-Service (SaaS) cloud infrastructure utilizing web application frameworks, RESTful APIs, and 256-bit SSL/TLS encryption for secure data transmission and storage. The main advantages of MyCase stem from its intuitive client portal, which streamlines client-practitioner communication, reduces scheduling friction, and simplifies routine file access. Conversely, its main disadvantages include its subscription-based pricing model—which creates recurring cost barriers for resource-constrained regional practices—and its lack of a public-facing attorney verification directory or localized court document archiving schemas. In relation to the proposed study, MyCase demonstrates how integrated web portals simplify appointment management and document organization, highlighting the operational necessity of pairing secure back-end case archives with accessible client communication channels in a localized platform.

## 2.4 Conceptual Framework

The conceptual framework of this study is structured using the Input-Process-Output (IPO) model to illustrate the systematic development, architectural implementation, and operational mechanics of the proposed Attorney Directory and Case File Archive System. Rather than treating these stages as isolated components, the framework demonstrates a continuous workflow where raw institutional data and policy requirements serve as the foundation, full-stack software engineering processes transform these resources, and the resulting web platform addresses operational inefficiencies in legal record management.

**INPUT**

- Physical Case Files & Metadata
- Attorney Profiles & Bar Credentials
- User Roles & Access Hierarchy
- System Specifications & Hardware Specs
- Regulatory Frameworks (RA 10173 DPA)

**PROCESS**

- Requirements Gathering & Analysis
- Database Normalization & System Architecture

• Full-Stack Development & Cloud Integration

• Cross-Module Integration & Security Rules

• Alpha/Beta Testing & ISO 25010 Evaluation

**OUTPUT**

• Integrated Directory & Case Archive System

• Cloud-Synchronized Digital Document Storage

• Metadata Indexing & Priority Search Engine

• Public Attorney Verification Portal

• Accelerated Retrieval & Security Compliance

**The Relationship Between Input and Process**

The Input stage establishes the foundational resources, domain requirements, and compliance parameters required to drive system development. These inputs directly inform every phase of the development process:

- **Physical Legal Records and Metadata:** Case numbers, filing dates, hearing events, and document types serve as raw data inputs that dictate the relational database schema, necessitating 3NF

normalization during the design process to eliminate operational redundancy.

- **Attorney Profiles and Credentials:** Verified bar details and specializations form the dataset required to build the public lookup and client-attorney matching mechanisms.

- **User Access Levels:** Roles defined for System Administrators, Attorneys, Secretaries, and Public Clients dictate the implementation of Role-Based Access Control (RBAC) protocols during full-stack coding.

- **System Specifications:** Software and hardware requirements guide the selection of backend frameworks, database management tools, and web hosting environments.

- **Legal Compliance (RA 10173):** Statutory mandates under the Data Privacy Act of 2012 establish strict security constraints, ensuring that encryption and access control logic are embedded throughout the development lifecycle.

**The Relationship Between Process and Output**

The Process stage converts raw inputs into a structured software application through a disciplined software development lifecycle (SDLC):

- **Requirements Analysis:** Synthesizes user access levels and legal compliance standards into concrete system specifications.

- **System and Database Design:** Translates physical document attributes into normalized database tables, establishing strict entity relationships and RBAC boundary controls.

- **System Development:** Implements full-stack web application code, integrates cloud storage infrastructure, and builds metadata indexing engines to handle physical file uploads.

- **Module Integration:** Connects the public-facing Attorney Directory with the restricted Case File Archive, ensuring seamless cross-module interaction without exposing private records.

- **System Testing and Quality Evaluation:** Subjects the platform to alpha/beta user testing, security auditing, and ISO 25010 quality assessments to verify system stability, usability, and operational speed.

**The Resulting System Output**

The culmination of these interconnected inputs and processes yields the final output: a fully realized web-based Attorney Directory and Case File Archive System. By processing physical files into normalized digital registries, the system delivers a cloud-synchronized repository that prevents document decay and misplacement. By applying RBAC and metadata indexing to user inputs, the system produces an advanced priority tracking dashboard and search engine for legal staff. Finally, by integrating public attorney profiles within privacy-compliant boundaries, the output establishes a secure directory portal—measurably accelerating file retrieval speeds, lowering transaction friction for clients, and safeguarding institutional legal data over time.

## 2.5 Synthesis of Related Literature and Studies

The reviewed literature, empirical studies, and existing platforms collectively demonstrate that digital transformation, normalized database architectures, and web-accessible portals are vital for modernizing legal administrative workflows. Centralizing document repositories and replacing manual filing logs with electronic case management systems (CMS) are consistently highlighted as necessary interventions to eliminate record degradation, reduce latency, and enforce institutional accountability across legal and judicial bodies (Bañez, 2025; Skipanes et al., 2026).

### Operational Bottlenecks and Literature Contradictions

A dominant thematic convergence across both local and international research is the severe operational friction caused by paper-dependent registries. Investigations by James et al. (2025) and Peñalba and Macabeo (2022) establish that physical record-keeping leads to unindexed files, severe case disposition delays, and administrative gridlock. To resolve these friction points, empirical evaluations confirm that migrating to web-based repositories eliminates retrieval delays and accelerates case processing (Acoba, 2024; Innovations for Poverty Action, 2023).

However, a critical comparison of the literature reveals a notable divergence in scope and technical priority between operational efficiency and security compliance. While studies focusing on workflow performance (Dela Gente et al., 2022; Antenor & Pomperada, 2025) prioritize rapid

document retrieval and user convenience, technical security literature (Guttman et al., 2022) emphasizes that unconstrained digital access introduces severe vulnerabilities, such as silent data corruption, metadata loss, and unauthorized record tampering. Furthermore, while national policy frameworks establish macro-level mandates for e-governance and judicial innovation (DICT, 2023; NEDA, 2023; Supreme Court of the Philippines, 2022), they offer limited practical architectural guidance for localized, resource-constrained law practices that must balance strict data privacy mandates—such as Republic Act No. 10173—with open public service delivery.

**Critical Analysis of Existing Systems and Architectural Limitations**

An analysis of existing legal information systems highlights why current software solutions fail to provide an end-to-end operational answer for local legal environments:

**1. Institutional Judicial Platforms (eCourt PH):** Government-driven platforms like eCourt PH (Supreme Court of the Philippines, 2024) are designed exclusively for formal court dockets and judicial personnel. Because these systems are restricted to state court dockets, independent legal practitioners, local law firms, and the public cannot utilize them to manage private case archives, track client-attorney interactions, or verify local practitioner profiles.

**2. Commercial Legal Practice Management Tools (Clio & MyCase):** Comprehensive commercial platforms like Clio (Themis Solutions, 2024) and MyCase (AffiniPay, 2025) offer robust internal tools for document management, billing, and private client portals. However, these platforms operate under commercial, profit-driven business models that require high, recurring per-user subscription fees. This creates financial barriers for small regional firms, legal aid organizations, and independent practices operating on limited budgets.

**3. Absence of Public-Facing Verification Directories:** Crucially, commercial practice management tools operate as closed, internal firm networks. They completely omit public-facing practitioner verification lookup tools. This structural limitation leaves prospective clients reliant on unverified third-party

information or physical inquiries when searching for licensed attorneys, creating information asymmetry and transaction friction prior to formal legal engagement (Munaweerage, 2025; Punitha et al., 2024).

Consequently, current practice leaves legal administration fragmented: organizations are forced to choose between expensive commercial tools that lack public verification capabilities, closed state court registries that exclude independent practices, or fragmented physical file systems that cause administrative backlogs.

**The Unresolved Research Gap and Proposed Solution**

The primary research gap identified across literature, empirical studies, and existing platforms is the absence of an affordable, integrated web platform that combines a public-facing attorney verification lookup with a secure, normalized back-end case file archive specifically tailored for local administrative standards.

The proposed Attorney Directory and Case File Archive System directly fills this gap. By unifying a public credential lookup portal, automated document indexing, role-based access control (RBAC), and cloud-synchronized file archiving into a single web application, the system eliminates paper-reliant backlogs, enforces data privacy compliance, and bridges the accessibility gap between legal practitioners and the public.

# CHAPTER III

# TECHNICAL BACKGROUND

## 3.1 Software Development Methodology

The study adopts the **Kanban Framework under the Agile Methodology** to develop the Attorney Directory and Case File Archive System. Rather than forcing rigid, fixed-length sprint cycles, Kanban allows the development team to focus on continuous, visual workflow management, limiting work-in-progress (WIP) to keep development steady and bottleneck-free.

Work is organized and tracked using a digital Kanban Board (via GitHub Projects / Trello) broken down into four clear continuous stages: Backlog, In Progress, Testing/Review, and Completed (Done). System requirements are converted into modular task cards, assigned to specific team members, and moved across the board as work progresses.

- **Work-in-Progress (WIP) Limits:** To prevent multitasking and maintain focus, a strict WIP limit of two active tasks per developer is enforced in the In Progress lane. A team member must test and complete their current task before pulling a new card from the backlog.
- **Task Synchronization & Review:** The team conducts frequent task check-ins to re-align board priorities, clear technical blockers, and review completed features as soon as they reach the *Testing/Review* stage.
- **Definition of Done (DoD):** A task card is only moved to *Completed (Done)* when it fulfills all the following criteria:

1. Source code is tested locally, refactored, and pushed to the main Git repository.
2. Database tables, foreign key constraints, and relational schemas execute cleanly without errors.
3. Form inputs, authorization logic, and backend endpoints pass functional testing.
4. Uploaded files successfully store to cloud storage (AWS S3) with verified database path references.

## 3.2 Requirements Analysis

Requirements analysis identifies the needs and expectations of the users of the Attorney Directory and Case File Archive System. It focuses on the functions needed to manage attorney information and organize case files digitally. The system is designed to reduce the difficulty of manually searching, storing, and maintaining attorney and case records.

### 3.2.1 Stakeholders

**Public Users / Unauthenticated Visitors** - Can search the public attorney directory and view sanitized, non-confidential case summaries without access to private files.

**Attorneys** - Authenticated users who manage assigned case files, review client details, upload documents, and track hearing schedules.

**Office Staff** - Authenticated users responsible for entering data, organizing case records, scheduling hearings, and retrieving archived files.

**System Administrator** - Manages user accounts, enforces role-based access control, configures lookup values, monitors audit logs, and oversees system backups.

**Law Office Management** - Monitors overall firm statistics, case statuses, and attorney directory records.

**Clients (External Beneficiaries)** - Benefit indirectly from streamlined case handling and faster document retrieval provided by attorneys and office staff.

### 3.2.2 Functional Requirements

**User Authentication & Role Access:** Allows authorized users to log in securely using hashed credentials and enforces role-based access permissions for System Administrators, Attorneys, and Office Staff.

**Attorney Directory & Management:** Stores, displays, and allows authorized users to manage attorney records (including contact info, bar credentials, and specializations) while providing a public directory view for unauthenticated visitors.

**Client Profile Management:** Enables authorized users to create, update, search, and deactivate client profiles, linking client records directly to their respective case files.

**Case File Management & Lifecycles:** Enables users to create, search, update, and manage case records. Case Status (Active, Pending, Closed, Archived) and Priority (Low, Medium, High, Urgent) can only be updated by authorized Attorneys and Staff, with all status changes requiring a transition reason logged to the audit trail.

**Case Assignment Management:** Allows Staff and Administrators to assign primary and co-counsel attorneys to cases. Reassignments preserve historical assignment records in the database rather than purging past data.

**Hearing Schedule Management:** Allows Attorneys and Office Staff to schedule, update, and manage court hearings linked to active cases, including hearing dates, times, court locations, and hearing types.

**Hearing Conflict Prevention:** Enforces automated schedule validation upon submission to prevent booking overlapping court hearings for the same attorney within a 1-hour time window.

**Digital File Archiving & Upload Controls:** Allows authorized users to upload, categorize, and store digital copies of legal documents associated with specific case records. System enforces strict file validation, limiting uploads to allowed formats (PDF, DOCX, PNG, JPG), imposing a **25MB maximum file size limit**, and restricting sensitive attachments based on role-based access permissions.

**Document Versioning:** Automatically generates incremental version numbers when an existing file is updated. Historical file versions remain immutable (read-only), while the system automatically marks the highest version number as active.

**Search & Advanced Metadata Filtering:** Enables users to search across cases, attorneys, and documents using keywords and advanced metadata filters (filing date, case category, priority, assigned attorney).

**Secure File Retrieval:** Restricts access and download capabilities for archived legal documents exclusively to verified internal roles.

**Public Non-Confidential Case Lookup:** Enables unauthenticated public visitors to search and view sanitized, non-confidential case summaries without exposing sensitive client details or private legal documents.

**System Lookups Management:** Allows Administrators to manage system-wide lookup tables, including case categories, priority levels, hearing types, and status lifecycles.

**Audit Trail & System Logging:** Automatically records mandatory system activities (logins, record modifications, document uploads/downloads) with user IDs and timestamps. Audit logs are append-only, strictly immutable, and accessible only to Administrators.

**Automated Cloud Backup & Recovery:** Conducts daily automated database snapshots and continuous binary file syncs to encrypted AWS S3 cloud storage, allowing Administrators to perform point-in-time restorations and enforce 7-year legal data retention policies.

### 3.2.2.1 Business Rules & System Constraints

To ensure data integrity, legal compliance, and reliable execution, the system enforces strict backend business logic across core modules:

- **Case Lifecycle & State Transitions:**
  - **Sequential Lifecycle Progression:** Case dockets must transition through explicit sequential states: **ACTIVE → PENDING → CLOSED → ARCHIVED**. Direct state jumping (for example, moving directly from ACTIVE to ARCHIVED) is strictly blocked by backend validation.
  - **Closure Validation Safeguard:** A case file cannot be updated to CLOSED or ARCHIVED if there are unresolved court hearings (status set to SCHEDULED or POSTPONED in the hearing schedule) or pending file upload tasks.
  - **Mandatory Transition Audit:** Every case status change requires a transition reason string, which is automatically recorded in the system audit log alongside the user ID and execution timestamp.
- **Automated Hearing Schedule Conflict Prevention:**
  - **1-Hour Attorney Time Buffer:** When a new court hearing is scheduled, the system automatically checks the assigned attorney's calendar.
  - **Overlap Logic Constraint:** The system blocks booking if the assigned attorney already has another hearing scheduled within 1 hour before or after the requested start time.

    - **Override Authorization:** Only System Administrators can bypass scheduling conflicts by marking a hearing as POSTPONED or assigning co-counsel to handle the overlapping proceeding.
- **Attorney Assignment & Historical Preservation:**
    - **Counsel Reassignment Safeguard:** Reassigning a case creates a new assignment record with an updated assignment date. Past counsel relationships are retained as historical records rather than purged or overwritten.
    - **Inactive Status Guard:** The system checks attorney account status prior to assignment. Inactive attorneys cannot be assigned to new or existing case files.
- **Document Immutability & Version Control:**
    - **Read-Only Historical Snapshots:** Modifying or updating an existing file entry increments the current version number and writes a read-only historical record to the document version table.
    - **Atomic Cloud Upload Sync:** File uploads execute as atomic transactions. If cloud storage fails, the database entry rolls back automatically to prevent orphaned records.

### 3.2.3 Non-Functional Requirements

**Performance & Responsiveness:** Page load times and query search results must render in under 2 seconds under normal operating conditions, supporting up to 50 concurrent active users without performance degradation.

**System Availability:** The system infrastructure must maintain 99.9% operational uptime (excluding scheduled off-peak maintenance).

**Security & Session Management:** User passwords must be hashed using bcrypt (cost factor 12). Inactive user sessions automatically terminate after 15 minutes. Data in transit is encrypted via HTTPS/TLS 1.3, and off-site cloud storage enforces AES-256 encryption at rest.

**Scalability & Capacity:** The database and storage architecture are designed to scale smoothly up to 10,000 active case records and 50,000 legal document files while maintaining performance benchmarks.

**Disaster Recovery:** In the event of system failure, automated cloud recovery must achieve a Recovery Time Objective (RTO) of under 4 hours and a Recovery Point Objective (RPO) of under 24 hours.

**Confidentiality & Data Access Control:** The system enforces strict Role-Based Access Control (RBAC) to separate public, internal, and confidential records as defined below:

| Data Category | Public Visitor View | Assigned Attorney View | Office Staff & Admin View |
|---|---|---|---|
| Attorney Data | Full Name, Specialization, Contact Info, Office Address, Aggregated Case Stats | Full Profile + Assigned Case List | Full Profile + Account Credentials & Role Controls |
| Client Profiles | **Hidden** (No Access) | Read-Only (For clients assigned to their cases) | Full Access (Create, Edit, Update, Deactivate) |
| Case Metadata | Sanitized Summaries / Public Dockets Only | Full Case Details (Assigned Cases Only) | Full Case Details (All System Cases) |
| Case Documents | **Hidden** (No Access) | View, Upload, and Version Assigned Case Files | Full Access (Upload, Archive, Manage All Files) |

### 3.3 Business Process Analysis

Business Process Analysis examines the existing process of managing attorney information and case files. It identifies the problems in the current manual process and explains how the proposed Attorney Directory and Case File Archive System can improve record management.

### 3.3.1 Current (As-Is) Process

Currently, legal records, attorney details, client profiles, and active case files are managed using physical folders, filing cabinets, and manual registers. Office staff manually collect client details during intake, record case details by hand, and physically transport dockets to assigned attorneys. When hearing notices arrive, dates are handwritten into physical desk calendars without centralized conflict checking. Searching for precedent files or updating active dockets requires staff to manually search through physical storage areas, presenting significant risks of misplaced files, unrecorded modifications, and compromised document confidentiality.

**Current Process Workflows:**

- **Case Filing & Retrieval Workflow:** Attorney/Client Intake → Information Collection → Manual Form Recording → Physical Folder Assignment → Filing Cabinet Storage → Physical Search & Retrieval → Manual Updating → Refiling Folder
- **Hearing & Scheduling Workflow:** Receive Court Notice → Manual Verification of Attorney Availability → Handwritten Entry in Desk Calendar → Physical Notification to Assigned Counsel → Post-Hearing Manual Docket Update
- **Access Control & Archiving Workflow:** Physical Request for Case File → Manual Authorization Check → Physical File Hand-Off → Manual File Return → Physical Storage in Secured Room

### 3.3.2 Proposed (To-Be) Process

The proposed system replaces manual paper processing with an integrated digital cloud platform. Office staff create digital **client profiles** and initialize case dockets with **case assignment** rules for primary and co-counsel. Document uploads are automatically tracked via **document versioning**, encrypted, and stored in secure cloud storage while writing actions to an immutable **audit logging** trail. **Hearings** are managed with automated conflict checking, and **role-based access control (RBAC)** restricts confidential legal data to authorized staff while providing a public-facing **attorney directory** search.

**Proposed System Workflows:**

- **Digital Case & Document Management Workflow:** Staff/Attorney Login → Client Profile Creation/Update → Case Initialization & Counsel Assignment → Validated Document Upload → Automated Versioning & Encryption → Audit Log Recording
- **Integrated Hearing Management Workflow:** Schedule Hearing Request → Automated System Conflict Check → Store Hearing Details → Automated Calendar Sync & Alert Notification → Post-Hearing Status Update
- **Role-Based Access & Public Search Workflow:** User Authentication Check → Role Evaluation (Public / Staff / Attorney / Admin) → Public

Attorney Directory & Non-Confidential Search OR Confidential File Access → Audit Trail Logging

### 3.3.3 Process Exception Paths

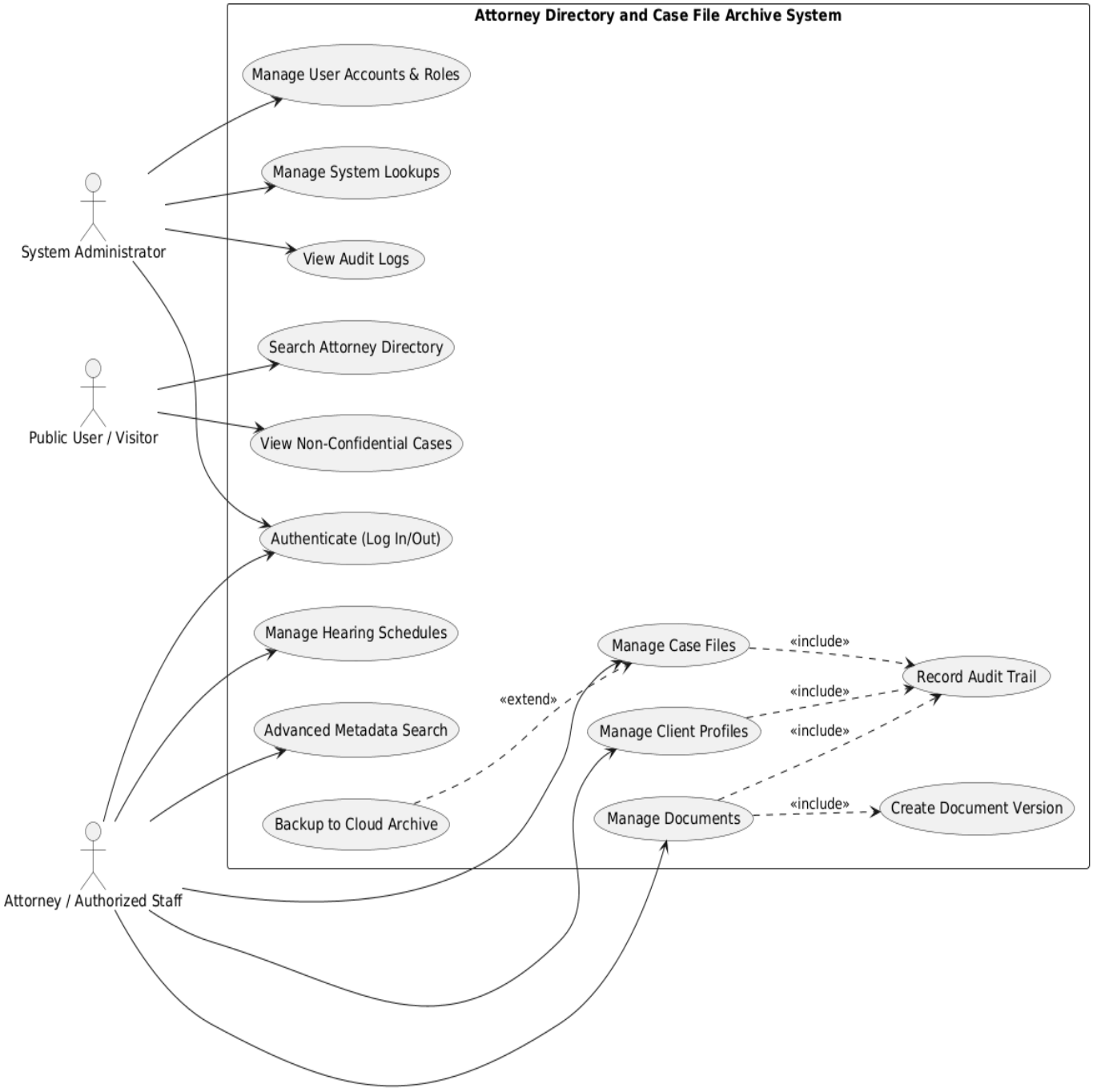
To ensure comprehensive system reliability and meet evaluation standards, the To-Be system defines explicit exception workflows for all error states and system edge cases:

- **Unauthorized Access Attempt:** User Requests Restricted Resource → RBAC Access Check Fails → Access Denied Notification Displayed → Security Exception Audit Log Created
- **Missing / Rejected Files:** User Uploads File → System Validation Fails (Format/Size/Corrupted) → File Rejected → Error Prompt Displayed to User with Guidelines
- **Duplicate Records:** User Submits New Record / Client Profile → System Scans Existing Database for Matching Identifiers → Match Found → System Blocks Creation & Prompts User to View Existing Record
- **Inactive Attorneys:** Staff Attempts Case Assignment → System Checks Attorney Status (is_active flag) → Attorney Marked Inactive → System Rejects Assignment & Prompts Selection of Active Counsel
- **Case Closure Workflow Exception:** Staff Requests Case Closure → System Validates Pending Tasks / Open Hearings → Unresolved Items Detected → System Blocks Closure and Lists Remaining Requirements

- **Document-Version Conflicts:** Concurrent Document Uploads Detected → System Compares Version Timestamps / Conflict Flags → Version Conflict Triggered → System Prompts Manual Merge or Save as New Alternate Version
- **Failed Backups / Cloud Sync Timeouts:** Cloud Upload or Backup Attempt → Network/API Timeout Occurs → System Retries (3x Automatic Attempts) → Failure Persists → Queue for Background Sync & Flag Admin Alert

## 3.4 System Design

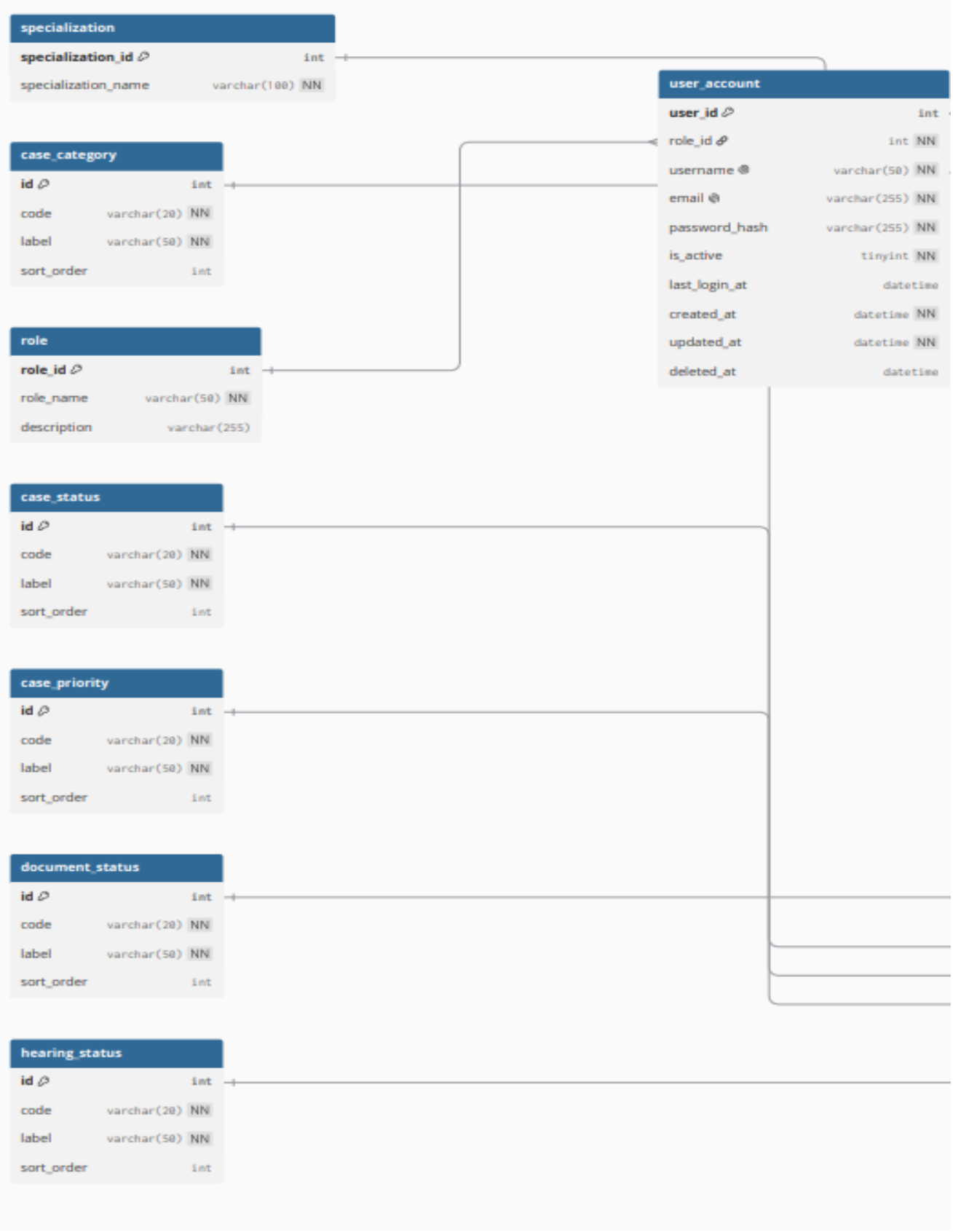
### 3.4.1 Use Case Diagram

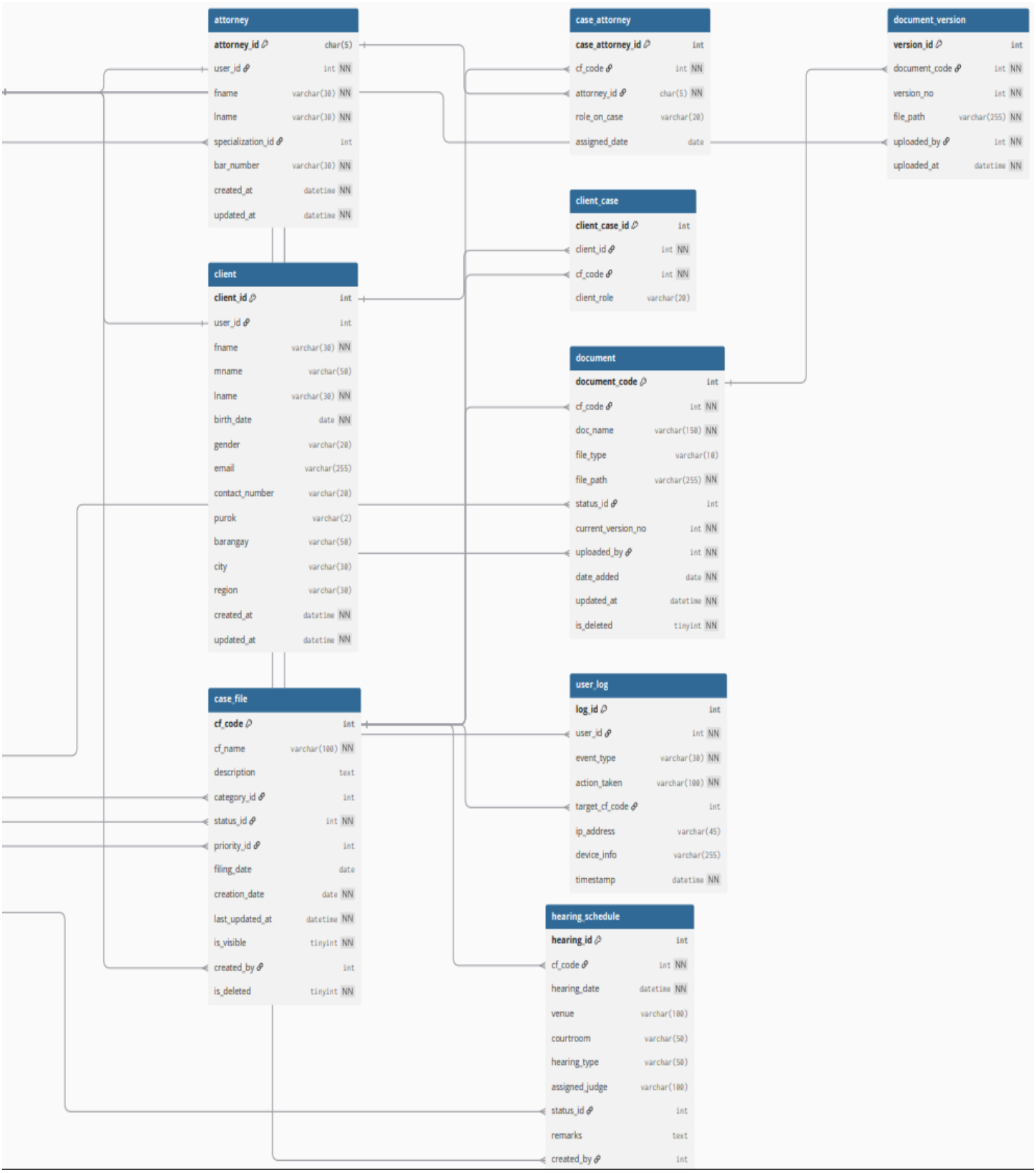



The use case model utilizes standard UML semantics to structure functional dependencies. Mandatory system operations—such as

enforcing non-repudiation via Record Audit Trail during case and document management and ensuring document history via Create Document Version—are modeled using <<include>> relationships. Optional or auxiliary system execution paths are modeled using <<extend>> relationships, specifically where Backup to Cloud Archive extends core case management actions conditionally upon storage transactions.

### 3.4.2 Entity Relationship Diagram




attorney
attorney_id char(5)
user_id int NN
fname varchar(30) NN
lname varchar(30) NN
specialization_id int
bar_number varchar(30) NN
created_at datetime NN
updated_at datetime NN
case_attorney
case_attorney_id int
cf_code int NN
attorney_id char(5) NN
role_on_case varchar(20)
assigned_date date
document_version
version_id int
document_code int NN
version_no int NN
file_path varchar(255) NN
uploaded_by int NN
uploaded_at datetime NN
client_case
client_case_id int
client_id int NN
cf_code int NN
client_role varchar(20)
client
client_id int
user_id int
fname varchar(30) NN
mname varchar(50)
lname varchar(30) NN
birth_date date NN
gender varchar(20)
email varchar(255)
contact_number varchar(20)
purok varchar(2)
barangay varchar(50)
city varchar(30)
region varchar(30)
created_at datetime NN
updated_at datetime NN
document
document_code int
cf_code int NN
doc_name varchar(150) NN
file_type varchar(10)
file_path varchar(255) NN
status_id int
current_version_no int NN
uploaded_by int NN
date_added date NN
updated_at datetime NN
is_deleted tinyint NN
user_log
log_id int
user_id int NN
event_type varchar(30) NN
action_taken varchar(100) NN
target_cf_code int
ip_address varchar(45)
device_info varchar(255)
timestamp datetime NN
case_file
cf_code int
cf_name varchar(100) NN
description text
category_id int
status_id int NN
priority_id int
filing_date date
creation_date date NN
last_updated_at datetime NN
is_visible tinyint NN
created_by int
is_deleted tinyint NN
hearing_schedule
hearing_id int
cf_code int NN
hearing_date datetime NN
venue varchar(100)
courtroom varchar(50)
hearing_type varchar(50)
assigned_judge varchar(100)
status_id int
remarks text
created_by int

***Database Normalization & Relational Structure***

To ensure data integrity, eliminate redundancy, and prevent update/deletion anomalies, the relational database structure is fully normalized up to Third Normal Form (3NF):

- First Normal Form (1NF): All table attributes contain single, atomic values. Multi-valued fields (such as multiple attorneys assigned to a single case) are broken out into dedicated junction tables (case_attorney). Primary keys are explicitly defined for every table.
- Second Normal Form (2NF): Every non-key attribute is fully functionally dependent on the entire primary key, eliminating partial dependencies. Composite join tables (like case_attorney and client_case) store relationship-specific attributes directly dependent on the composite key.
- Third Normal Form (3NF): All transitive dependencies have been removed by extracting lookup and reference data into separate lookup entities (role, specialization, case_category, case_status, case_priority, document_status, and hearing_status). Foreign keys strictly reference these normalized lookup tables.

### 3.4.3 Database Schema

1. **role**

| Field Name | Data Type | Key | Nullability | Description & Access Control |
|---|---|---|---|---|
| role_id | INT | PK | NOT NULL | Unique primary key for user roles. |
| role_name | VARCHAR(50) | | NOT NULL | System role title (Admin, Attorney, Staff). |
| description | VARCHAR(255) | | NULL | Role permissions summary. |

2. **specialization**

| Field Name | Data Type | Key | Nullability | Description & Access Control |
|---|---|---|---|---|
| specialization_id | INT | PK | NOT NULL | Unique identifier for area of practice. |
| specialization_name | VARCHAR(100) | | NOT NULL | Title of specialization [Public Directory]. |

### 3. case_category

| Field Name | Data Type | Key | Nullability | Description & Access Control |
|---|---|---|---|---|
| id | INT | PK | NOT NULL | Unique identifier for case category. |
| code | VARCHAR(20) | | NOT NULL | Classification code (CIVIL, CRIMINAL, ADMIN). |
| label | VARCHAR(50) | | NOT NULL | Display label for case category. |
| sort_order | INT | | NULLABLE | UI display priority sequence. |

4. **case_status**

| Field Name | Data Type | Key | Nullability | Description & Access Control |
|---|---|---|---|---|
| id | INT | PK | NOT NULL | Unique identifier for case lifecycle status. |
| code | VARCHAR(50) | | NOT NULL | System code (ACTIVE, PENDING, CLOSED, ARCHIVED). |
| label | VARCHAR(255) | | NOT NULL | UI display label for status. |
| sort_order | INT | | NULLABLE | UI display priority sequence. |

5. **case_priority**

| **Field Name** | **Data Type** | **Key** | **Nullability** | **Description & Access Control** |
|---|---|---|---|---|
| id | INT | PK | NOT NULL | Unique identifier for priority level. |
| code | VARCHAR(50) | | NOT NULL | Urgency code (LOW, MEDIUM, HIGH, URGENT). |
| label | VARCHAR(255) | | NOT NULL | UI display label for priority. |
| sort_order | INT | | NULLABLE | UI display priority sequence. |

### 6. document_status

| Field Name | Data Type | Key | Nullability | Description & Access Control |
|---|---|---|---|---|
| id | INT | PK | NOT NULL | Unique identifier for document state. |
| code | VARCHAR(50) | | NOT NULL | State code (ACTIVE, SUPERSEDED, ARCHIVED). |
| label | VARCHAR(255) | | NOT NULL | UI display label for document status. |
| sort_order | INT | | NULLABLE | UI display priority sequence. |

### 7. hearing_status

| Field Name | Data Type | Key | Nullability | Description & Access Control |
|---|---|---|---|---|
| id | INT | PK | NOT NULL | Unique identifier for court hearing status. |
| code | VARCHAR(50) | | NOT NULL | Status code (SCHEDULED, COMPLETED, POSTPONED, CANCELLED). |
| label | VARCHAR(255) | | NOT NULL | UI display label for hearing status. |
| sort_order | INT | | NULLABLE | UI display priority sequence. |

8. **user_account**

| Field Name | Data Type | Key | Nullability | Description & Access Control |
|---|---|---|---|---|
| user_id | INT | PK | NOT NULL | Primary key identifier for system user account. |
| role_id | INT | FK | NOT NULL | Foreign key referencing role.role_id. |
| username | VARCHAR(50) | | NOT NULL | Unique account login username. |
| email | VARCHAR(255) | | NOT NULL | Registered system contact email. |
| password_hash | VARCHAR(255) | | NULLABLE | Salted password hash generated via bcrypt. |
| is_active | TINYINT(1) | | NOT NULL | Account active flag (1=Active, 0=Disabled). |
| last_login_at | DATETIME | | NULLABLE | Audit timestamp of last user authentication. |
| created_at | DATETIME | | NOT NULL | Account registration timestamp. |
| updated_at | DATETIME | | NOT NULL | Account profile modification timestamp. |
| deleted_at | DATETIME | | NULLABLE | Soft-delete timestamp for account deactivation. |

9.

**10. attorney**

| Field Name | Data Type | Key | Nullability | Description & Access Control |
|---|---|---|---|---|
| attorney_id | CHAR(5) | PK | NOT NULL | Unique primary key for attorney record. |
| user_id | INT | FK | NOT NULL | Foreign key referencing user_account.user_id. |
| fname | VARCHAR(30) | | NOT NULL | First name [Public Directory / Internal Full]. |
| lname | VARCHAR(30) | | NOT NULL | Last name [Public Directory / Internal Full]. |
| specialization_id | INT | FK | NULLABLE | Foreign key referencing specialization.specialization_id [Public Directory]. |
| bar_number | VARCHAR(30) | | NULLABLE | Official legal Roll of Attorneys bar number [Internal / Admin Only]. |
| created_at | DATETIME | | NOT NULL | Profile creation timestamp. |
| updated_at | DATETIME | | NOT NULL | Profile update timestamp. |

**11. client**

| Field Name | Data Type | Key | Nullability | Description & Access Control |
|---|---|---|---|---|
| client_id | INT | PK | NOT NULL | Primary key identifier for client profile. |
| user_id | INT | FK | NOT NULL | Foreign key referencing user_account.user_id. |
| fname | VARCHAR(30) | | NOT NULL | First name [Confidential - Assigned Attorney / Staff / Admin Only]. |
| mname | VARCHAR(50) | | NULLABLE | Middle name [Confidential - Assigned Attorney / Staff / Admin Only]. |
| lname | VARCHAR(30) | | NOT NULL | Last name [Confidential - Assigned Attorney / Staff / Admin Only]. |
| birth_date | DATE | | NOT NULL | Birth date [Confidential - Assigned Attorney / Staff / Admin Only]. |
| gender | VARCHAR(20) | | NULLABLE | Gender identity string. |
| email | VARCHAR(255) | | NULLABLE | Primary email [Confidential - Assigned Attorney / Staff / Admin Only]. |
| contact_number | VARCHAR(20) | | NULLABLE | Contact number [Confidential - Assigned Attorney / Staff / Admin Only]. |

| Field Name | Data Type | Key | Nullability | Description & Access Control |
|---|---|---|---|---|
| purok | VARCHAR(2) | | NULLABLE | Address sub-unit [Confidential - Assigned Attorney / Staff / Admin Only]. |
| barangay | VARCHAR(50) | | NULLABLE | Barangay designation [Confidential - Assigned Attorney / Staff / Admin Only]. |
| city | VARCHAR(30) | | NULLABLE | Municipality/City name [Confidential - Assigned Attorney / Staff / Admin Only]. |
| region | VARCHAR(30) | | NULLABLE | Region designation [Confidential - Assigned Attorney / Staff / Admin Only]. |
| created_at | DATETIME | | NOT NULL | Profile creation timestamp. |
| updated_at | DATETIME | | NOT NULL | Profile update timestamp. |

**12. case_file**

| Field Name | Data Type | Key | Nullability | Description & Access Control |
|---|---|---|---|---|
| cf_code | INT | PK | NOT NULL | Primary key for legal case docket. |
| cf_name | VARCHAR(100) | | NOT NULL | Official case title [Public Docket Sanitized]. |
| description | TEXT | | NULLABLE | FCase background details [Confidential - Internal Only]. |
| category_id | INT | FK | NULLABLE | Foreign key referencing case_category.id. |
| status_id | INT | FK | NOT NULL | Foreign key referencing case_status.id [Public Docket Sanitized]. |
| priority_id | INT | FK | NULLABLE | Foreign key referencing case_priority.id [Internal Only]. |
| filing_date | DATE | | NULLABLE | Court filing date [Public Docket Sanitized]. |
| creation_date | DATE | | NOT NULL | Initial system logging date. |
| last_updated_at | DATETIME | | NOT NULL | Timestamp of last docket update. |
| is_visible | TINYINT(1) | | NOT NULL | Public visibility flag (1=Public, 0=Private). |

| Field Name | Data Type | Key | Nullability | Description & Access Control |
|---|---|---|---|---|
| created_by | INT | FK | NULLABLE | Foreign key referencing user_account.user_id of logger. |
| is_deleted | INT | | NOT NULL | Soft-deletion flag (0=Active, 1=Deleted). |

### 13. case_attorney

| Field Name | Data Type | Key | Nullability | Description & Access Control |
|---|---|---|---|---|
| case_attorney_id | INT | PK | NOT NULL | Unique junction table record ID. |
| cf_code | INT | FK | NOT NULL | Foreign key referencing case_file.cf_code. |
| attorney_id | CHAR(5) | FK | NOT NULL | Foreign key referencing attorney.attorney_id. |
| role_on_case | VARCHAR(20) | | NULLABLE | Counsel role designation (Lead Counsel, Co-Counsel). |
| assigned_date | DATE | | NULLABLE | Formal assignment date. |

### 14. client_case

| Field Name | Data Type | Key | Nullability | Description & Access Control |
|---|---|---|---|---|
| client_case_id | INT | PK | NOT NULL | Unique junction table record ID. |
| client_id | INT | FK | NOT NULL | Foreign key referencing client.client_id. |
| cf_code | INT | FK | NOT NULL | Foreign key referencing case_file.cf_code. |
| client_role | VARCHAR(20) | | NULLABLE | Party role (Plaintiff, Defendant, Petitioner). |

### 15. document

| Field Name | Data Type | Key | Nullability | Description & Access Control |
|---|---|---|---|---|
| document_code | INT | PK | NOT NULL | Primary key for legal file entry. |
| cf_code | INT | FK | NOT NULL | Foreign key referencing parent case_file.cf_code. |
| doc_name | VARCHAR(150) | | NOT NULL | File display title [Internal Only]. |
| file_type | VARCHAR(10) | | NULLABLE | File format (PDF, DOCX, PNG, JPG). |
| file_path | VARCHAR(255) | | NOT NULL | AWS S3 cloud storage object reference key. |
| status_id | INT | FK | NULLABLE | Foreign key referencing document_status.id. |
| current_version_no | INT | | NOT NULL | Pointer to latest active version (default 1). |
| uploaded_by | INT | FK | NOT NULL | Foreign key referencing user_account.user_id of uploader |
| date_added | DATE | | NOT NULL | System upload date. |
| updated_at | DATETIME | | NOT NULL | Timestamp of last file edit. |
| is_deleted | TINYINT(1) | | NOT NULL | Soft-deletion flag (0=Active, 1=Deleted). |

### 16. document_version

| Field Name | Data Type | Key | Nullability | Description & Access Control |
|---|---|---|---|---|
| version_id | INT | PK | NOT NULL | Primary key for historical file version instance. |
| document_code | INT | FK | NOT NULL | Foreign key referencing parent document.document_code. |
| version_no | INT | | NOT NULL | Incremental version index (1, 2, 3...). |
| file_path | VARCHAR(255) | | NOT NULL | Immutable S3 storage object path for historical version. |
| uploaded_by | INT | FK | NOT NULL | Foreign key referencing user_account.user_id of submitter. |
| uploaded_at | DATETIME | | NOT NULL | Immutable timestamp when this version was committed. |

### 17. hearing_schedule

| Field Name | Data Type | Key | Nullability | Description & Access Control |
|---|---|---|---|---|
| hearing_id | INT | PK | NOT NULL | Primary key for scheduled court hearing. |
| cf_code | INT | FK | NOT NULL | Foreign key referencing target case_file.cf_code. |
| hearing_date | DATETIME | | NOT NULL | Date and time of court proceeding. |

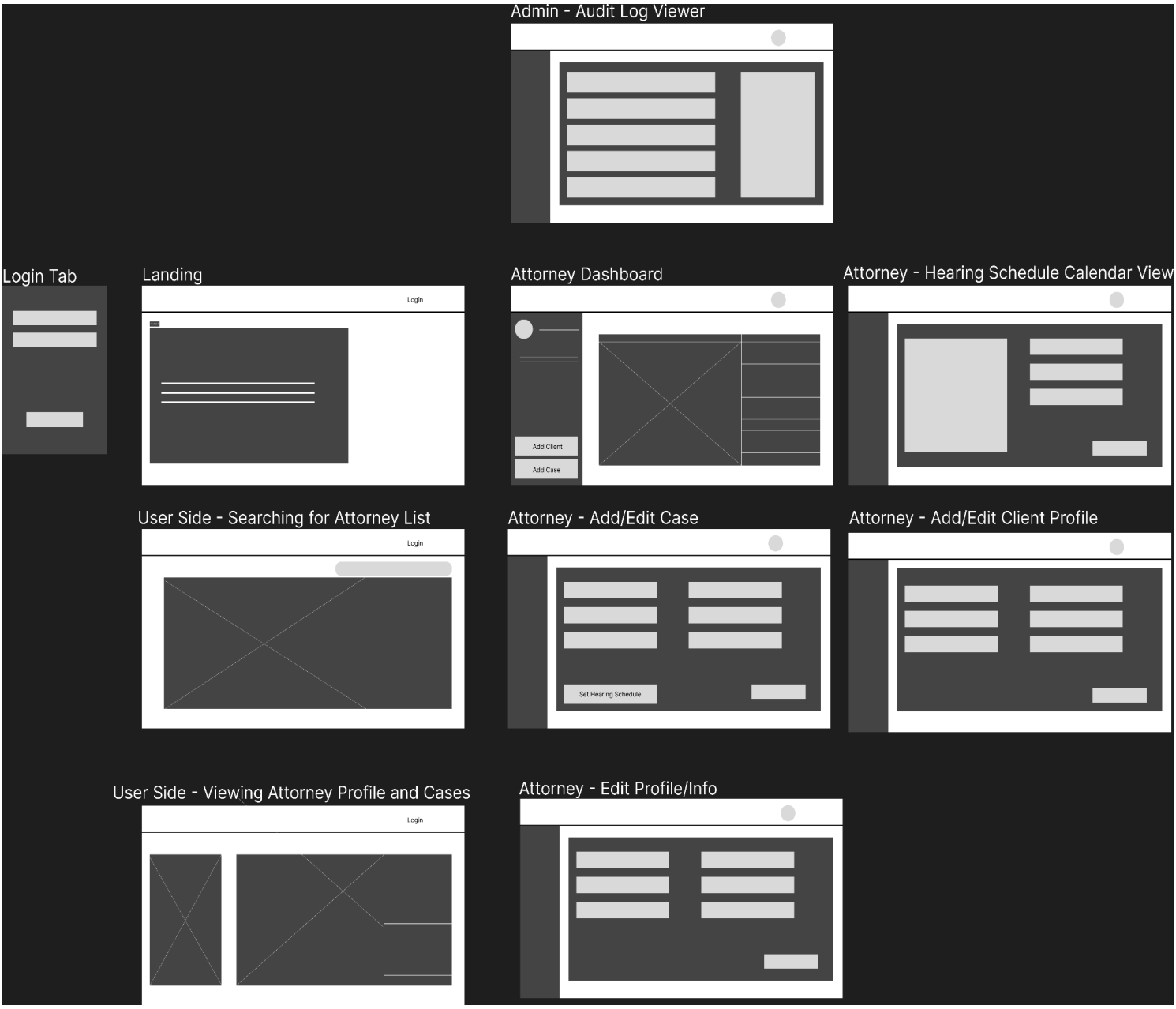
| Field Name | Data Type | Key | Nullability | Description & Access Control |
|---|---|---|---|---|
| venue | VARCHAR(100) | | NULLABLE | Court address/branch location. |
| courtroom | VARCHAR(50) | FK | NULLABLE | Designated courtroom designation. |
| hearing_type | VARCHAR(50) | | NULLABLE | Type of proceeding (Arraignment, Pre-Trial, Trial). |
| assigned_judge | VARCHAR(100) | | NULLABLE | Presiding judge name. |
| status_id | INT | FK | NULLABLE | Foreign key referencing hearing_status.id. |
| remarks | TEXT | | NULLABLE | Post-hearing outcome details or calendar notes. |
| created_by | INT | FK | NULLABLE | Foreign key referencing user_account.user_id of scheduler. |
| created_at | DATETIME | | NOT NULL | Entry creation timestamp. |

**18. user_log**

| Field Name | Data Type | Key | Nullability | Description & Access Control |
|---|---|---|---|---|
| log_id | INT | PK | NOT NULL | Primary key for system audit trail entry. |
| user_id | INT | FK | NOT NULL | Foreign key referencing user_account.user_id of actor. |
| event_type | VARCHAR(30) | | NOT NULL | Event classification (AUTH, CREATE, |

| Field Name | Data Type | Key | Nullability | Description & Access Control |
|---|---|---|---|---|
| | | | | UPDATE, DELETE, DOWNLOAD). |
| action_taken | VARCHAR(100) | | NOT NULL | Action detail string executed. |
| target_cf_code | INT | FK | NULLABLE | **Foreign key referencing affected case_file.cf_code**. |
| ip_address | VARCHAR(45) | | NULLABLE | Client IP address string. |
| device_info | VARCHAR(255) | | NULLABLE | Client browser user-agent header. |
| timestamp | DATETIME | | NOT NULL | Execution execution timestamp. |

### 3.4.4 Wireframe



## 3.5 Project Tools and Technologies

**Frontend: HTML5, CSS3, & Modern JavaScript (Vanilla ES6+)**

- Delivers the client-side user interface without extra framework overhead. Handles DOM manipulation, dynamic form submissions, and layout rendering for public directory searches and admin panels.

**Backend: Node.js / Express.js**

- Runs the server-side runtime and RESTful API framework. Handles incoming HTTP requests, executes business logic, manages role-based access, and coordinates data flow between the web interface and database. Implements JSON Web Tokens (JWT) for secure session authentication and Multer middleware to process and validate multipart file uploads (restricting formats to PDF/DOCX up to 25 MB).

**Database: MySQL**

- Provides structured, ACID-compliant relational storage. Enforces data normalization and relational integrity across client profiles, attorney directory metadata, hearing schedules, and system audit trails. User account passwords are securely hashed using bcrypt prior to database insertion.

**Modeling & Documentation: PlantUML / PlantText**

- Compiles Diagram-as-Code scripts into standardized, version-controlled UML models (including Use Case diagrams and ERDs) directly alongside the application code.

**Version Control: Git & GitHub**

- Manages source code versioning, tracks technical revisions, and facilitates feature branch management during system development.

**Cloud Storage & Hosting: AWS S3 / Cloud Storage**

- Manages secure, off-site binary object storage for physical document uploads. Enforces S3 Bucket Versioning to preserve historical document revisions, AES-256 Server-Side Encryption, and private bucket access via temporary time-limited signed URLs generated by the backend API. Implements automated daily snapshot backups, soft-deletion retention rules, and a 7-year lifecycle retention policy for legal document archiving prior to permanent disposition.
- MySQL & Cloud File Consistency: To maintain consistency, MySQL stores a unique object key (s3_object_key) mapped to the document ID rather than raw binary data. File uploads execute as atomic transactions: if a cloud upload fails, the database transaction rolls back; if the database save fails, the uploaded file is immediately purged to prevent orphaned data. Deletions execute a soft-delete in MySQL (is_deleted = TRUE) while cloud storage archives historical versions.

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